WG.48.1: Independence of Textual Order of Sections

Proposal for the Z Standard, by Wolfgang Grieskamp (wg@cs.tu-berlin.de), November 11, 1998, (v2)

Note This is a technical refinement of an earlier proposal by SHV.

What

The order how sections appear in a Z specification document is arbitrary. The parent-relation of sections must be acyclic.

Why

Shortly: the adequate way to structure a specification document with multiple sections is not always bottom-up. (As far as I know, the panel already agreed that section reordering is useful in general.)

How

Conformance Tools working in batch mode which don't implement this feature are considered still as conforming, since filters can be build which perform the section reordering for them.

Updates to WD 1.5 Type Inference Rules, Clause 13.2.1.1 (the according clause in 12, resp.): add a comment such as:

> The order how sections appear in a Z specification document is arbitrary. The parent-relation of sections must be acyclic.

Introduce reordering (and cycle check) as a condition to the rule. This could be something along:

$$\frac{\begin{array}{c}\{\}\ \bullet^{s}\ s_{\rho(1)} : \Gamma_{\rho(1)} \\ \dots \\ \{i_{\rho(1)} \mapsto \Gamma_{\rho(1)}, \dots, i_{\rho(n\Gamma 1)} \mapsto \Gamma_{\rho(n\Gamma 1)}\}\ \bullet_{s}\ s_{\rho(n)} : \Gamma_{\rho(n)}\end{array}}{\bullet^{Z}\ s_1 \dots s_n} \quad (\rho : 1 \,..\, n \rightarrowtail\!\!\!\!\!\!\prec\ 1 \,..\, n)$$

It's sufficient to just demand that the re-ordering ρ is a bijection, since:

- any instance of ρ which does not represent a proper bottom-up order is rejected by rule 13.2.2.1, which would detect that a parent reference i_k is not in the environment build from previous sections in the order
- if there are degrees of freedom how to reorder sections, then because of independency of individual sections, so that the chosen order doesn't matters

Implementation Tools can easily recognise the syntax of section paragraphs already at scanning time. So a three-phase architecture of parsing can be implemented: lexical analysis, section reordering (by topological sorting), and syntax analysis.